

Resilient Provenance Tracking for IoT Cold Chains via Logical Clocks

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1 Background & Problem Statement

1.1 The Operational Vulnerability

Biologics, mRNA vaccines, and temperature-sensitive pharmaceuticals require an unbroken cold chain (typically strictly regulated between 2°C and 8°C, or ultra-cold thresholds at -70°C) from the point of origin to clinical administration. A thermal excursion lasting even a few hours can degrade protein structures and silently destroy vaccine potency without altering physical appearance or packaging integrity. Concurrently, globalized pharmaceutical supply chains suffer from the penetration of counterfeit or adulterated medications, exploiting visibility gaps across fragmented logistical links.

Both challenges stem from a single systemic defect: siloed enterprise custody logging. The primary logistical handlers- Manufacturer, Distributor, Cold-Storage Hub, and Dispensing Pharmacy- record physical environmental conditions and custody transfers on disparate, private relational databases. Reconstructing the end-to-end chain of custody requires querying each entity sequentially, forcing downstream consumers to place blind institutional trust in the honesty and record retention of each upstream intermediary.

1.2 The Distributed Systems Breakdown

Transitioning cold-chain verification to a replicated, distributed ledger resolves cross-enterprise data isolation, but exposes a deeper synchronization failure: **the invalidity of physical wall-clock time across edge devices experiencing intermittent connectivity**.

Internet-of-Things (IoT) edge dataloggers attached to shipment packaging pass through connectivity blackouts- including inter-state transit corridors, cargo holds, and shielded cold-storage vaults. During these blackouts, sensors buffer telemetry locally and upload historical batches upon establishing a network link.

If the underlying distributed ledger sequences these buffered readings using physical wall-clock or GPS timestamps, **genuine causal misorderings occur by construction**:

- Hardware oscillator drift (inherent in low-power crystal oscillators subjected to cryogenic cycles) introduces drift vectors (Δt).
- Network reconnection latency (δ_{sync}) delays the broadcast of transit departure events until after an on-grid warehouse has already registered an arrival event.
- A warehouse's **Received** event can obtain a lower physical timestamp than a carrier's buffered **Departed** event, or vice versa.

This temporal distortion corrupts the audit trail. A critical temperature spike occurring in transit can be causally misplaced after delivery confirmation, falsely assigning civil or regulatory

liability to the receiving facility while absolving the negligent logistics carrier. Conversely, an excursion that occurred prior to handoff may be masked altogether if the overlapping, out-of-order logs fail to establish which entity exercised physical custody when the violation transpired.

2 Project Objectives

1. **Design a Permissioned Ledger Architecture:** Implement a deterministic, multi-party ledger representing four logistical roles (Manufacturer, Distributor, Cold Storage, Pharmacy) using cryptographically linked append-only blocks.
2. **Implement a Vector-Clock Causal Ordering Engine:** Design an event-ordering layer that abandons physical timestamps in favor of Fidge-Mattern vector clocks, guaranteeing zero causal inversions for all buffered sensor and custody events across arbitrary connectivity gaps.
3. **Develop a Matrix-Clock Autonomous Log-Pruning Protocol:** Construct a secondary matrix-clock mechanism tracking mutual knowledge horizons across all network nodes, enabling localized, garbage collection of intermediate telemetry logs without consensus communication.

3 System Architecture & Methodology

The system architecture follows a four-stage data pipeline designed for intermittent connectivity:

1. **IoT Sensor Node (In Transit):** Operates at the edge. It tracks physical parameters (Temperature, GPS), maintains a local Vector Clock (V_{device}), and buffers generated events locally during RF connectivity blackouts.
2. **Causal Merge Engine:** Triggered upon network reconnection. It verifies cryptographic signatures and applies happens-before partial ordering to deterministically sequence buffered events into the global timeline.
3. **Permissioned Ledger State:** A 4-node distributed network (Manufacturer, Distributor, Cold Storage, Pharmacy) that maintains the replicated, causally-ordered log of all shipments.
4. **Matrix Clock Pruning Engine:** Operates via periodic state exchanges. It evaluates the global knowledge horizon to autonomously prune obsolete telemetry from active memory without requiring consensus communication.

3.1 Data Schema & Ledger Primitives

Instead of physical timestamps, the ledger utilizes two immutable, cryptographically signed event structures driven by logical clocks:

- **SensorEvent:** Contains the shipment ID, telemetry data (e.g., temperature, coordinates), and a snapshot of the IoT device's current Vector Clock.
- **CustodyEvent:** Records the transfer of custody between network entities, containing the sender/receiver IDs and the merged Vector Clock representing the handoff.

3.2 Causal Ordering & Autonomous Pruning

Let $N = \{M, D, C, P\}$ represent the fixed set of $n = 4$ institutional network entities.

Vector-Clock Causal Ordering:

Every participant maintains a vector clock V of dimension n .

- Local updates: $V_i[i] \leftarrow V_i[i] + 1$
- State merges: $V_j[k] \leftarrow \max(V_j[k], V_{msg}[k]), \quad \forall k \in \{1, \dots, n\}$

When an edge node reconnects and uploads its buffer, the engine topologically sorts incoming payloads according to the strict happens-before relation ($e_a \rightarrow e_b$), ensuring correct chronological insertion regardless of how far the hardware clocks drifted during the blackout.

Matrix-Clock Log Pruning (Garbage Collection):

To prevent ledger state-bloat on resource-constrained nodes, the system deploys Matrix Clocks (M_i) to track mutual knowledge. Element $M_i[j, k]$ denotes node i 's knowledge of the latest event at node k that has been causally observed by node j .

An intermediate **SensorEvent** can be safely pruned from a node's active memory if and only if the column-minimum condition is met:

$$\min_{j \in \{1, \dots, n\}} M_i[j, k] \geq V(e_{sensor})[k], \quad \forall k \in \{1, \dots, n\}$$

This condition mathematically proves that every valid node in the network has causally incorporated the event, allowing it to be safely discarded without risking state divergence.

4 Evaluation & Metrics

The system will be evaluated using a discrete-event simulation harness built in C++, subjecting the logistics chain to synthetic yet empirically-grounded transit distortions (heavy clock drift and blackout distributions). The evaluation is designed around two primary quantifiable benchmarks:

4.1 Quantification of Causal Violations

To measure the causal failure rate, the experimental harness will execute two parallel pipelines consuming an identical event stream:

1. **Causal-Order Pipeline:** Evaluates the zero-violation guarantee and state consolidation using the proposed vector/matrix clock implementation.
2. **Naive Physical-Timestamp Baseline:** Orders events strictly using scalar hardware timestamps generated by simulated edge system clocks subjected to drift (± 100 ms/hr up to ± 10 s/day) and transmission latency jitter.

The performance delta between these pipelines is captured by the following metrics:

- **Causal Misordering Rate (R_{err}):** Defined as the ratio of inverted event pairs (e_a, e_b) where $e_a \rightarrow e_b$ but $T_{ledger}(e_b) < T_{ledger}(e_a)$, plotted as a function of average offline duration:

$$R_{err} = \frac{\sum \mathbb{I}(T(e_b) < T(e_a) \mid e_a \rightarrow e_b)}{\text{Total Event Pairs}}$$

- **False Liability Attribution Rate (F_{liab}):** Frequency with which a temperature excursion occurring under the physical custody of node k is falsely assigned to node $k + 1$ due to logging inversions around transfer windows.

5 Scope Boundaries & Constraints

- **Hardware Attestation Excluded:** The ledger guarantees the temporal ordering and causal consistency of reported data. Physical sensor spoofing and hardware-level security are considered external infrastructure and remain out of scope.
- **Linear Topology Scoping:** To strictly isolate synchronization behavior, the simulation models a linear supply chain ($M \rightarrow D \rightarrow C \rightarrow P$). Complex supply chain branching is managed logically via shipment IDs rather than complex graph joins.
- **Bounded Scalability Overhead:** While vector and matrix clocks introduce $O(N^2)$ message complexity, this overhead is trivial in this context. Pharmaceutical cold chains utilize permissioned consortia with limited enterprise roles ($N \leq 10$), avoiding the scaling bottlenecks of public networks.

6 References

1. L. Lamport, “Time, Clocks, and the Ordering of Events in a Distributed System,” *Communications of the ACM*, vol. 21, no. 7, pp. 558–565, 1978.
2. C. J. Fidge, “Timestamps in Message-Passing Systems that Preserve the Partial Ordering,” *Proceedings of the 11th Australian Computer Science Conference*, vol. 10, no. 1, pp. 56–66, 1988.
3. F. Mattern, “Virtual Time and Global States of Distributed Systems,” *Parallel and Distributed Algorithms*, pp. 215–226, 1989.
4. “An End-To-End Log Management Framework for Distributed Systems,” *IEEE Conference Publication*, IEEE Xplore.
5. K. M. Chandy and L. Lamport, “Distributed Snapshots: Determining Global States of Distributed Systems,” *ACM Transactions on Computer Systems*, vol. 3, no. 1, pp. 63–75, 1985.